MTH202 MIDSEM 1

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Note: All variables a, b, c, x, y, z, r should be taken to have values in integers.

Exercise 1 (4 points). Assume that $x \equiv a \pmod r$ and $y \equiv b \pmod r$, which of the following is True? Either prove (if True) or provide a counter-example (if False) to the following statements:

(1) $xy \equiv ab \pmod r$
(2) $x^y \equiv a^b \pmod r$

Exercise 2 (4 points). Which of the following equations in x and y have solutions in integers? Either prove that they do not have a solution or provide a solution:

(1) $1129x + 2267y = -1$
(2) $649x + 413y = 1$

Exercise 3 (4 points). Show that if

(1) 9 divides $(x^3 + y^3 + z^3)$, then 9 divides $(x^2y^2z^2)$.
(2) Find a positive integer that leaves a remainder of 1 when divided by 2, a remainder of 2 when divided by 3, a remainder of 3 when divided by 4, ..., and a remainder of 2024 when divided by 2025.

Exercise 4 (4 points). Answer the following:

(1) Describe the set of all positive integers x such that $x \equiv 0 \pmod{11}$ and $x \equiv 0 \pmod{37}$, and in particular find the minimum element in this set.
(2) Describe the set of all (not necessarily positive) integers y such that $y \equiv 7 \pmod{11}$ and $y \equiv 33 \pmod{37}$. In particular, find the integer y in this set such that $|y|$ is smallest possible.

Exercise 5 (4 points). Find the remainder when $2^{2^{2025}}$ is divided by 7.

"Can you do Addition?" the White Queen said. "What's one and one and one and one and one and one and one and one and one and one?" "I don't know," said Alice. "I lost count." "She can't do Addition," the Red Queen interrupted.
(From Alice in Wonderland. Here is to me hoping that all of us can do addition!).
